

# CAT User and Administrator Manual

A shared human and AI workspace for discussing a codebase



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# 1 Introduction

CAT is a small chat workspace in which a research group discusses a specific GitHub codebase, with AI agents participating as named members of the conversation. It resembles Slack in shape—a sidebar of channels, a message list, a composer—but it is deliberately much smaller, and it makes one commitment Slack does not: a channel is bound to a repository, and the agents in that channel can read it. The exception proves the rule: **#general** is bound to nothing, and is simply a room to talk in.

The system is designed for a group of fewer than twenty people. That constraint is not incidental. It is what makes it reasonable to let every member read every channel, to manage membership by adding rows to a table, and to run the whole thing on free hosting.

A second constraint shapes it just as much: only some people may invoke an AI agent. Everyone can read, everyone can talk, but summoning an agent spends money against an API key, and that permission is granted individually. Hiding a button would not be a control, so the check lives on the server, in the one place the key exists.

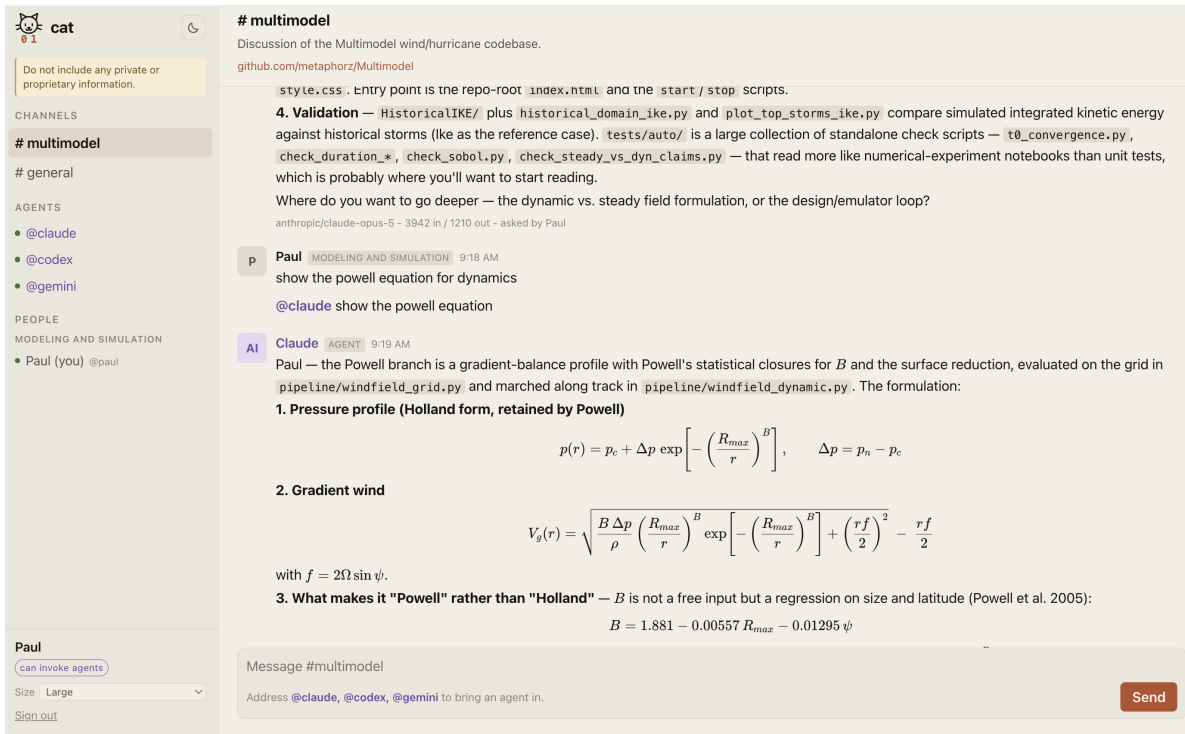


Figure 1: The CAT workspace, mid-conversation. Channels, agents and people are listed in the sidebar, the last grouped by discipline; the button beside the mark switches between light and dark. The agent answers *a particular reader*—it opens with the reader’s name, because it was told who asked and what they work on—and writes the wind field in the notation of the discipline rather than in prose, which the browser renders from the  $\text{\LaTeX}$  the model produced. The grey line beneath a reply records which model answered, what it cost, and who asked for it.

## 1.1 Design goals

- **One channel per codebase.** `#multimodel` is about `github.com/metaphorz/Multimodel`. Adding a repository means adding a row, not deploying anything. A channel with no repository named on it, such as `#general`, is a conversation rather than a codebase; the agent is told as much and drops the codebase framing.
- **Agents are participants, not features.** They appear in the roster, they are addressed by name, and their replies are ordinary messages in the transcript.
- **An interdisciplinary group is the normal case.** Members carry a discipline, and the agent is told who is asking, so a statistician and a meteorologist asking the same question get different answers.
- **Permissions are enforced where they cannot be bypassed,** and are separated by consequence: asking a question, spending money, and writing to a repository are three different decisions.
- **No build step.** The client is three static files. It is deployed by copying them to GitHub Pages.

## 1.2 What CAT is not

An agent can open a pull request when explicitly asked, on a channel whose repository permits it, by a person permitted to ask. It cannot merge one. Every change an agent makes arrives as a branch and a pull request for a human to review, and that review is the point rather than a formality—see Section 5.5.

There are no threads, reactions, file uploads, or message editing. Recreating Slack is not the goal; having somewhere to think about a codebase together is.

## 2 Architecture

CAT is a static web page and a managed backend. There is no server to administer and nothing to keep running.

**GitHub Pages** serves `index.html`, `app.js`, `style.css` and `config.js`. These are static files with no build step, published by a GitHub Actions workflow on every push.

**Supabase** provides authentication, a PostgreSQL database with row level security, a realtime change feed, and the Edge Function runtime. The free tier is far larger than this workload requires.

**The Edge Function** (`invoke-agent`) is the security boundary. It identifies the caller, decides whether they may summon an agent, and only then reaches for the API key.

**OpenRouter** reaches Anthropic, OpenAI and Google models through a single credential, which is what makes adding an agent a database row rather than a code change.

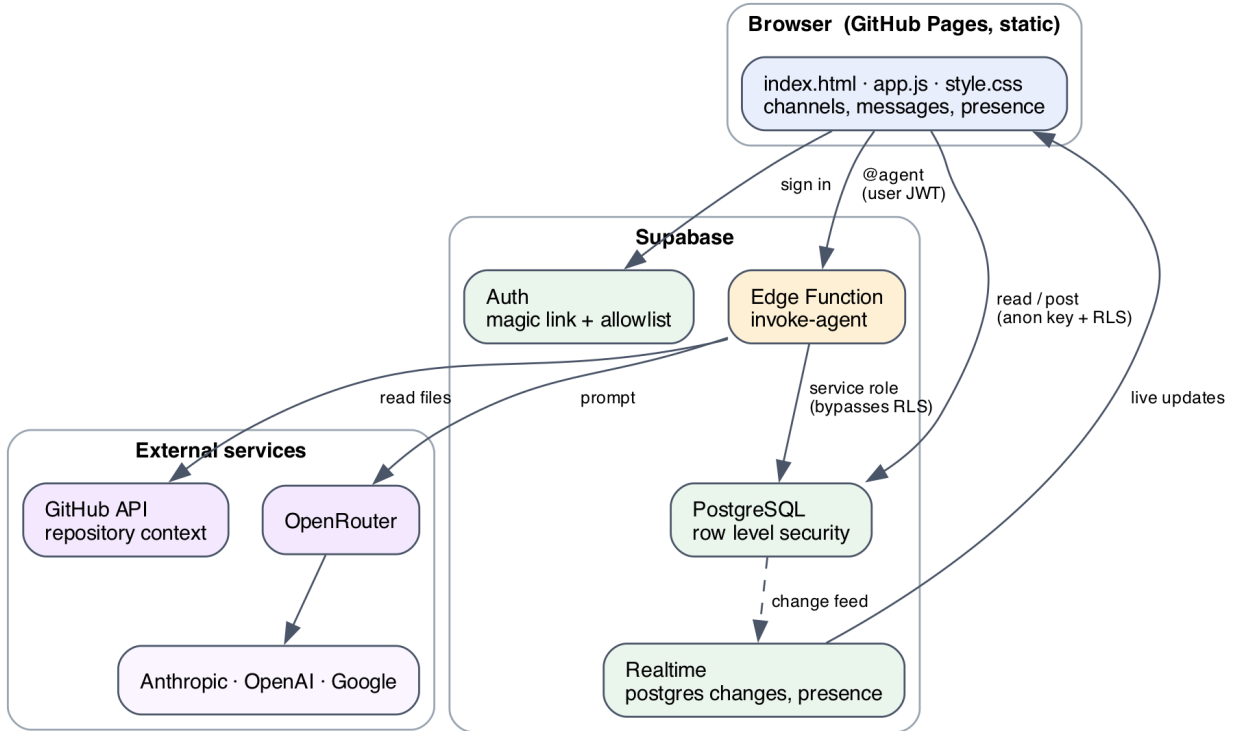


Figure 2: The whole system. The browser holds no secrets; the Edge Function holds all of them.

## 2.1 Why the browser can hold the keys it does

`config.js` contains the Supabase project URL and the publishable (anonymous) key, and both are visible to anyone who views source. This is intended. The anonymous key is an identifier, not an authorisation: every table is protected by row level security, and an unauthenticated caller holding that key can read nothing at all. The credentials that matter—the OpenRouter key, the GitHub token, the service role key—exist only as Edge Function secrets and never reach a browser.

## 2.2 Why the agent answers asynchronously

A model call on a real question can take longer than a browser will keep a request open. Rather than hold the connection, `invoke-agent` inserts an empty message with `status = 'pending'`, returns immediately, and finishes the work in the background, updating that row when the answer arrives.

The realtime feed turns this into a feature. Every connected client sees “Claude is thinking” the moment the agent is summoned, and sees the reply replace it when it lands—without polling and without reloading.

## 3 The permission model

Three mechanisms hold the system up, and it is worth knowing which does what, because they fail differently.

**Getting in at all** is the `allowlist` table together with a trigger on user creation. An address

that is not listed cannot create an account: the trigger raises an exception and the signup fails. This matters because the page is public—anyone can load the sign-in screen.

**Reading and posting** is row level security. A member may insert messages as themselves and as nobody else, and messages carrying an `agent_slug` are rejected outright, so a collaborator cannot forge words from an agent. Agent messages are written only by the service role, which the browser never possesses.

**Invoking an agent** is the Edge Function checking `can_invoke_agent` before it touches the API key.

|               | Read | Post | Invoke agents | Request changes |
|---------------|------|------|---------------|-----------------|
| Paul          | yes  | yes  | yes           | yes             |
| Other members | yes  | yes  | no            | no              |
| Agents        | yes  | yes  | —             | —               |
| Anyone else   | no   | no   | no            | no              |

Table 1: The default arrangement. Both agent permissions are granted per person in the `members` table, and both default to false.

The last column is deliberately not the same permission as the third. Invoking an agent spends money; asking one to change code also writes to a repository. Folding them together would mean that granting a collaborator the ability to ask a question also granted them the ability to commit.

A third control sits on the channel rather than the person: `channels.allow_writes`. A repository can be discussed without being writable, and a channel may only ever write to the repository it is bound to. The blast radius of the whole feature is therefore exactly the repositories named in channels that have been deliberately opened.

### 3.1 A subtlety worth stating

Row level security decides *which rows* a member may change; it cannot decide *which columns*. Members are allowed to edit their own profile row, and without a further restriction that would let anyone grant themselves `can_invoke_agent`. The schema therefore also issues a column-level grant:

```
revoke update on public.members from authenticated;
grant update (display_name, github_login, username, specialty)
  on public.members to authenticated;
```

`can_invoke_agent` and `is_admin` are deliberately absent from that list.

## 4 Data model

Seven tables, shown in Figure 5.

The distinction between `allowlist` and `members` is the one that confuses people. `allowlist` is an invitation: an address, and what that person will be allowed to do. `members` is created automatically the first time they actually sign in. Someone invited but absent exists in the first table and not the second, and is therefore invisible in the interface.

## 4.1 Specialties and code areas

Each member carries a **specialty**—meteorologist, statistician, actuary, engineer, geospatial analyst, computer scientist, and so on. Each channel additionally maps specialties to real paths in *that* repository, through **channel\_areas**. The mapping is per channel rather than global because “the statistics code” means something different in every repository.

The agent receives both. When a statistician asks how the wind field is generated, it can answer in the vocabulary of sampling and variance and point at `pipeline/fit_metamodels.py`; when a meteorologist asks the same question it can talk about surface roughness and point at `pipeline/windfield.py`. When a question lands squarely in someone else’s discipline, it is instructed to say so and name that person rather than answer at the edge of its competence.

## 4.2 Code requests

**code\_requests** records every pull request an agent has been asked to open: the brief, who asked, and how it ended—**dispatched**, **running**, **opened** or **failed**, with the branch and pull request URL once they exist. It is kept apart from **messages** because it has a lifecycle, and because it is the audit trail for everything the system has ever written to a repository. Every member can read it; only the service role writes it.

## 4.3 Messages

A message is written either by a person or by an agent, never both, and a check constraint enforces it:

```
constraint messages_speaker_check check (
  (author_id is not null and agent_slug is null) or
  (author_id is null and agent_slug is not null)
)
```

The **status** column carries **pending**, **complete** or **error**, which is how the “thinking” state is represented. The **metadata** column records, for agent messages, which model answered, who asked, and the token usage—visible in small type beneath each reply.

# 5 Using CAT

## 5.1 Signing in

There is no password. Enter an allowlisted email address, and a single-use link arrives by mail.

Two things commonly go wrong. The link must be opened on a machine that can reach the address it redirects to, and it must be opened in the browser you intend to use—a session created in one browser does not exist in another.

## 5.2 Channels

A channel usually corresponds to a repository, named for it, and the header links directly to it on GitHub. Selecting a channel loads its history and subscribes to live updates; the channel is reflected in the page URL, so a particular channel can be linked to or bookmarked.

**#general** is the one channel with no repository behind it. The header shows no link, the agent is told there is no codebase in front of it, and the instruction to keep the discussion on the code is replaced by permission to follow the conversation wherever it goes. Agents cannot open pull

requests there, since there is no repository to write to. It is meant to accumulate and be cleared; `tools/flush-channel.sh` writes the transcript to a local file and then deletes every message in the channel, leaving the channel itself in place.

### 5.3 Talking to an agent

Mention an agent by name anywhere in a message:

```
@claude what does this codebase model, and where should a
statistician look first?
```

The composer indicates whether the mention will summon anyone. Members without permission may still write `@claude`—the text is not forbidden—but nothing will answer, and the composer says so plainly rather than failing silently.

The agent is given the recent conversation, the roster with everyone’s discipline, the specialty-to-path map for that channel, and the repository’s overview document and source file listing. It is told who asked and what their discipline is.

### 5.4 Formatting, tables and mathematics

Messages are rendered as Markdown, and agents are told so. Fenced code blocks, tables with column alignment, headings, lists, block quotes and rules all render. Mathematics is written in  $\text{\LaTeX}$ —`$...$` inline and `$$...$$` displayed—and typeset with  $\text{\LaTeX}$ , so a wind field can be discussed in notation rather than in prose describing notation.

The renderer is written by hand rather than taken from a library. Message content is untrusted, so everything is escaped before any structure is applied to it; adopting a Markdown library would mean adopting somebody else’s sanitiser along with it.

### 5.5 Asking for a change

On a channel whose repository allows writes, someone permitted to request changes can simply ask:

```
@claude make a PR that samples cp and rmax dependently, the way
we just discussed
```

There is no special command word. Whether a pull request was actually requested is decided by the model, through a tool it is offered only when both permissions already allow it. This matters more than it might appear: “make a PR for that” and “we shouldn’t open a PR for this yet” differ by something only a reader of English can judge, and a keyword match would get the second one wrong.

What is dispatched is the brief *and the conversation*. A decision reached over twenty messages does not survive being restated in one sentence—the constraints, the alternatives considered, and the reasons for rejecting them live in the discussion. The coding agent receives both and is told to read both.

The job then works on a real checkout, so it can read the surrounding code and run the tests, rather than patching blind. It pushes a `cat/request-N` branch and opens a pull request; it never touches the default branch. The link is posted back into the channel, and so is any failure—a request that disappears silently from the conversation is worse than one that fails loudly.

Nothing merges itself.

## 5.6 Appearance

Two controls, both in the sidebar, both remembered per browser.

**Light or dark**, from the button at the top beside the mark. Light is the default and is Anthropic’s tan, #f0eee6, with the clay accent that goes with it; dark is the palette the interface was originally drawn in. The button shows the theme you would be switching *to* rather than the one you are in. Two details are deliberate: the saved choice is applied before the page paints, so choosing dark does not mean a flash of white on every load; and the operating system’s own light/dark setting is not consulted, so the same person does not find two different-looking workspaces on two machines.

**Size**, offering five steps. It scales the entire interface—type, avatars, sidebar width—rather than type alone, because a large font inside a fixed-width column reads worse than no change at all.

## 6 Administration

### 6.1 Slash commands

Most of the administration below can also be done from the composer. Typing / opens a palette of commands; the arrow keys move through it and ENTER completes the one you want. The palette appears only for members whose `is_admin` flag is set.

That last sentence describes the interface, not the protection. As everywhere else in CAT, hiding a control is cosmetic: the `admin-command` Edge Function re-checks `is_admin` on every call, so a member who reads `app.js` and posts the request by hand is refused by the server rather than by the absence of a menu.

| command              | what it does                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>/model</code>  | On its own, lists what each agent is running. With a search term, a shortlist of what that vendor currently offers—and since an agent name is also a search term, <code>/model claude</code> answers both readings: what <code>@claude</code> is on now, and what it could be moved to. With an agent and a model identifier, switches that agent’s default. |
| <code>/retry</code>  | Re-runs your last question to an agent on a different model, leaving the stored default alone.                                                                                                                                                                                                                                                               |
| <code>/status</code> | Message, token and cache counts by channel, person and model.                                                                                                                                                                                                                                                                                                |
| <code>/who</code>    | Everyone on the allowlist, and which of them have yet to sign in.                                                                                                                                                                                                                                                                                            |
| <code>/cost</code>   | OpenRouter credit remaining.                                                                                                                                                                                                                                                                                                                                 |
| <code>/invite</code> | Adds a row to the allowlist.                                                                                                                                                                                                                                                                                                                                 |

A command is never posted. It is intercepted in the composer before the `insert`, so there is no message row, no realtime event, and nothing for other members to see. This also keeps commands out of the transcript that `invoke-agent` hands the model as conversation history, which matters: a stored `/model` line would otherwise arrive in the next prompt as noise. The answer is rendered locally and is gone on the next reload.

Two of these deserve a note. `/model` changes stored state: the switch applies to every channel and every member, permanently, and nobody is told. `/retry` is the smaller instrument, and is usually the one wanted when a particular answer is simply taking too long. Both refuse any model that OpenRouter reports cannot use tools—roughly one in six of them—because the agent reads the repository through tool calls, and an agent that has silently lost that ability looks like a model that has become vague rather than one that cannot open a file. OpenRouter’s `:batch` variants are hidden for a similar reason: they are half price and asynchronous, so a switch to one would look like a bargain and then never answer.

What `/model claude` returns is a shortlist rather than the catalogue: OpenRouter carries fourteen tool-capable Anthropic models, most of them older versions of four lines. The command keeps the newest of each family and ranks them by output price, which is the nearest thing to a capability ordering the catalogue offers—it puts Opus above Sonnet above Haiku without the code having to know that those names exist, and it will go on doing so as the version numbers move. Six at most, which in practice today means four for Anthropic and three each for Google and the Codex line. `/model claude all` gives the unabridged list for the rare case that matters.

## 6.2 Adding a person

Either `/invite alice@example.edu statistics` Alice Reyes, or the row it writes, in the Supabase SQL editor:

```
insert into public.allowlist
(email, display_name, username, specialty, can_invoke_agent)
values
('alice@example.edu', 'Alice Reyes', 'alice', 'statistics', false);
```

They then sign in themselves; the member record is created on first login. Setting the specialty is worth doing at this point, since it is what lets the agent pitch its answers correctly.

To change an existing member’s permissions, update `members`—the `allowlist` value only seeds the initial state:

```
update public.members set can_invoke_agent = true
where username = 'alice';
```

## 6.3 Seeing who is in the workspace

Three states are easy to confuse: *invited* (on the allowlist, possibly never seen), *member* (has signed in at least once), and *present* (has the application open right now). Only the last two appear in the sidebar, and presence is live state that is not stored anywhere. `/who` (§6.1) answers the first two from the composer; the query behind it is:

```
select a.email,
       coalesce(m.display_name, a.display_name) as name,
       m.username,
       (m.id is not null) as has_signed_in,
       u.last_sign_in_at
from public.allowlist a
left join public.members m on lower(m.email) = lower(a.email)
left join auth.users u on u.id = m.id
order by u.last_sign_in_at desc nulls last, a.email;
```

## 6.4 Adding a channel

```
insert into public.channels
(slug, name, purpose, github_owner, github_repo, position)
values
('hurricane', 'hurricane', 'Storm_tracking_generation.',
'metaphorz', 'Hurricane', 1);
```

If the repository is private, the configured GitHub token needs read access to it as well. Consider also populating `channel_areas` for the new channel, since that is what lets the agent direct people to the right files.

Leaving `github_owner` and `github_repo` null makes a channel that is about nothing in particular, as `#general` is.

## 6.5 Adding an agent

Because the Edge Function dispatches on `agents.provider`, adding a model requires no deployment at all:

```
insert into public.agents (slug, display_name, provider, model)
values ('gemini', 'Gemini', 'openrouter',
'google/gemini-3.1-pro-preview');
```

Adding a new *provider*—something other than `anthropic` or `openrouter`—does require a branch in the function and a redeployment.

Note that adding agents widens what can be asked for; it does not widen who may ask.

# 7 Deployment

## 7.1 Supabase

Create a project, then run the migrations in `supabase/migrations/` in order, in the SQL editor. Under **Authentication** → **URL Configuration**, set the site URL and the redirect URLs, including both the published address and `http://localhost:8000/` for local work. Note the trailing slashes: the client asks to be returned to `origin + pathname`, which always ends in one, and an entry lacking it will not match.

The built-in email sender is limited to roughly two messages per hour and is intended for testing. Configure custom SMTP before inviting anyone.

## 7.2 The Edge Function

```
supabase secrets set --project-ref <ref> OPENROUTER_API_KEY=...
supabase secrets set --project-ref <ref> GITHUB_TOKEN=...
supabase functions deploy invoke-agent --project-ref <ref>
```

`SUPABASE_URL`, `SUPABASE_ANON_KEY` and `SUPABASE_SERVICE_ROLE_KEY` are injected automatically and must not be set by hand.

The GitHub token is optional. Without it the agent can still discuss the conversation, but a private repository is invisible to it; it is told so, and says so, rather than guessing.

### 7.3 Enabling pull requests for a repository

Three further pieces, and they are deliberately separate from the read path.

First, a *second* GitHub credential. Reading a repository and writing to one are different privileges and should not share a token. Use a fine-grained personal access token limited to the single repository, with **Contents** and **Pull requests** set to read and write, and nothing else. It is worth confirming the scope empirically rather than trusting the form: the token should return 404 for any other private repository you own.

```
supabase secrets set --project-ref <ref> GITHUB_WRITE_TOKEN=...
supabase secrets set --project-ref <ref> CAT_CALLBACK_SECRET=...
supabase functions deploy agent-callback --project-ref <ref> --no-verify-jwt
```

The callback runs without JWT verification because GitHub Actions holds no Supabase session; it authenticates with the shared secret instead. It is narrow by design—it can update one code request and post one message about it—so that the service role key stays in Supabase rather than being handed to GitHub.

Second, install `workflows/multimodel-cat-agent.yml` into the target repository at `.github/workflows/cat-` with three repository secrets: `OPENROUTER_API_KEY` for the coding agent, `CAT_CALLBACK_URL` and `CAT_CALLBACK_SECRET` for the return path. The coding agent bills through OpenRouter like the chat agents do, so the workspace spends from one balance rather than two; the workflow points the CLI at OpenRouter’s Anthropic-shaped endpoint through `ANTHROPIC_BASE_URL` and passes the key as `ANTHROPIC_AUTH_TOKEN`. Note that the write token above deliberately lacks Workflows permission, so the agent cannot modify its own CI; install the file with a separate credential.

Third, open the channel:

```
update public.channels set allow_writes = true where slug = 'multimodel';
update public.members set can_request_changes = true where username = 'paul';
```

Both default to false, so a repository is never writable by accident.

### 7.4 GitHub Pages

Pages must be served from a public repository unless the account has a paid plan. This is safe: no credential lives in any committed file. A public page is not public access—anyone may load the sign-in box, and the allowlist decides everything after that.

## 8 Testing

`tests/test_screens.py` drives a real browser through every screen using Selenium: the sign-in gate, the refusal of an address that is not on the allowlist, the signed-in workspace, posting, code-block rendering, escaping of markup in messages, and the agent hint in the composer. Screenshots are written to `tests/screenshots/`.

Two aspects are worth noting. Magic-link authentication leaves nothing to type, so the suite mints a sign-in link through Supabase’s admin API and drives the browser to it. And because there is only one database, everything the suite writes is tagged and deleted afterwards.

```
export SUPABASE_SERVICE_ROLE_KEY=...
python3 tests/test_screens.py -v # all but the agent
CAT_TEST_AGENT=1 python3 tests/test_screens.py -v # including it
```

Invoking an agent costs money, so that test is opt-in. By default the suite checks that the composer offers the agent, not that a vendor answers.

## 9 Limitations and future work

- **An agent can open a pull request but cannot review one.** It does not see CI results, reviewer comments, or whether the branch still merges cleanly a week later. The pull request is where its involvement ends.
- **The conversation is an input to a process that writes code.** On a writable channel, everything typed there may reach a job holding a write credential. With a single person permitted to request changes this is academic; it stops being academic the moment a second person is granted that permission.
- **Every member sees every channel.** Reasonable below twenty people; it would need per-channel membership beyond that.
- **Repository context is a file listing and an overview document,** not the code itself. The agent knows what exists and can reason about structure, but cannot read a function body it was not shown.
- **Free-tier projects pause after a week of inactivity** and are resumed from the dashboard with data intact.
- **Presence is not history.** There is no record of who was here yesterday, only who is here now and when each person last signed in.

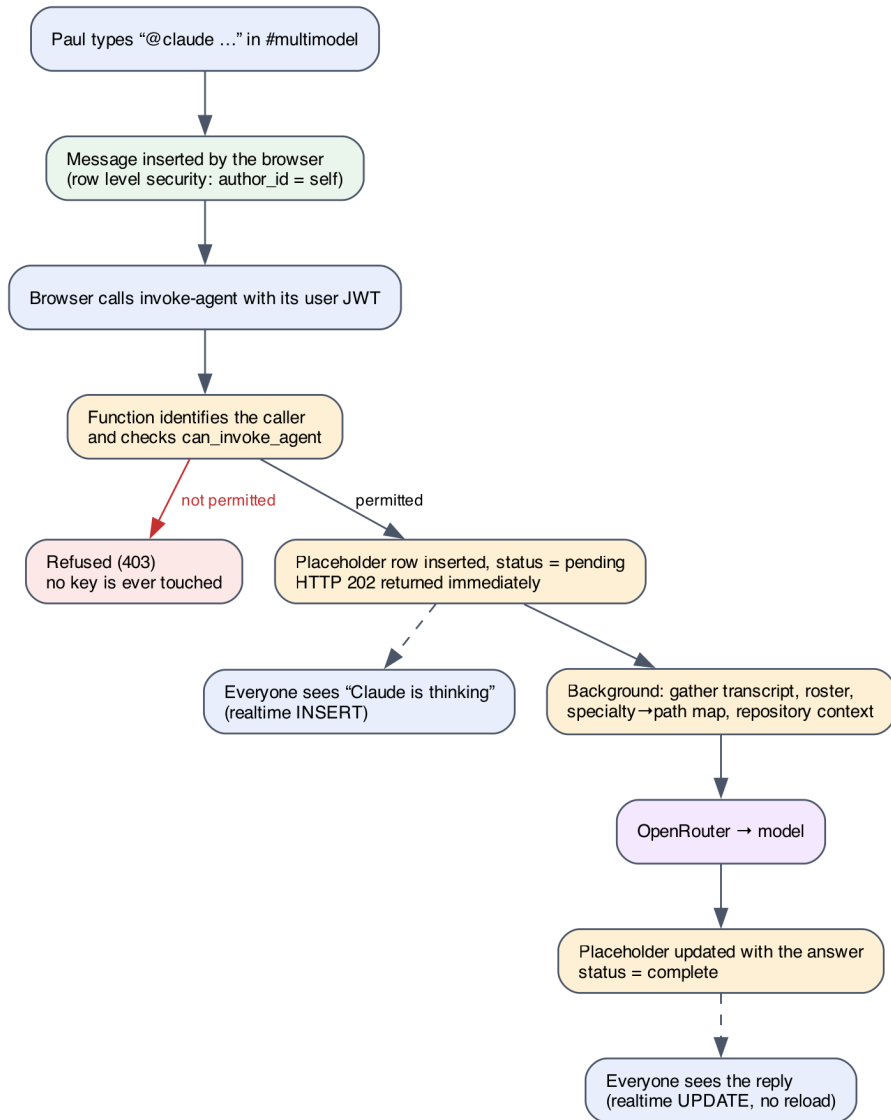


Figure 3: Invoking an agent. The permission check happens before any credential is touched, and the placeholder makes the wait visible to everyone.

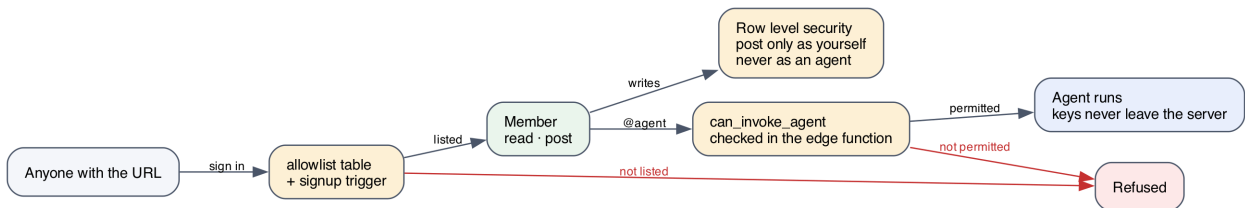


Figure 4: Where each decision is made. Nothing in the browser is trusted.

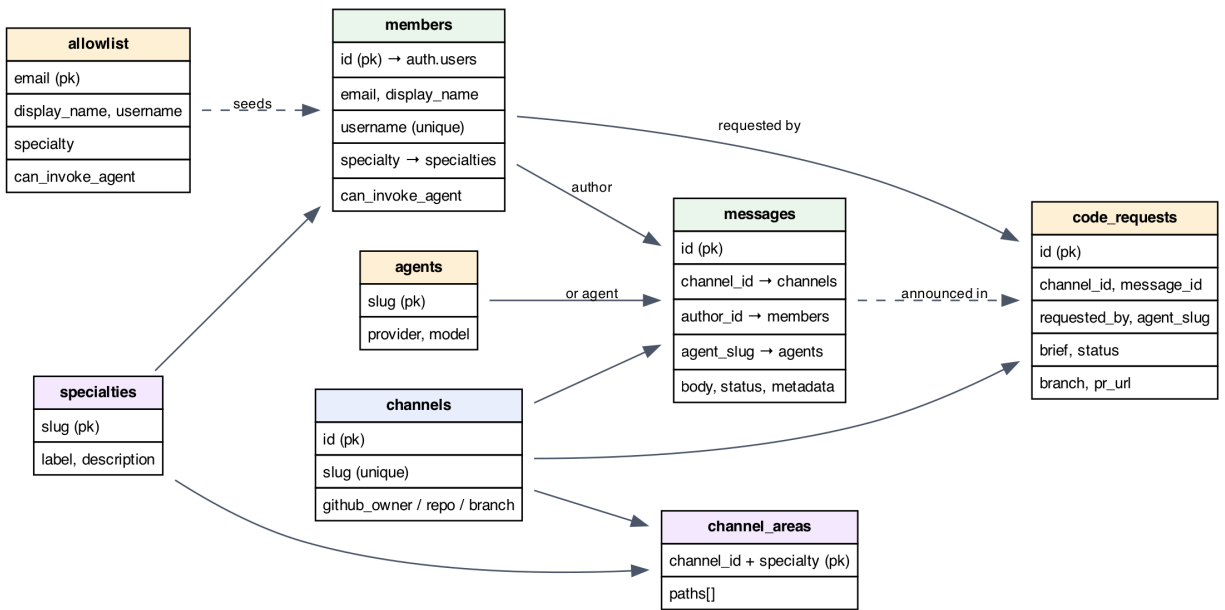


Figure 5: The schema. **allowlist** governs who may exist; **members** records who does.

cat

A shared workspace for talking about code.

Email

you@example.com

Send sign-in link

Access is by invitation. If your address is not on the workspace allowlist, no link will get you in.

Figure 6: The sign-in screen. It is all anyone sees without an invitation.

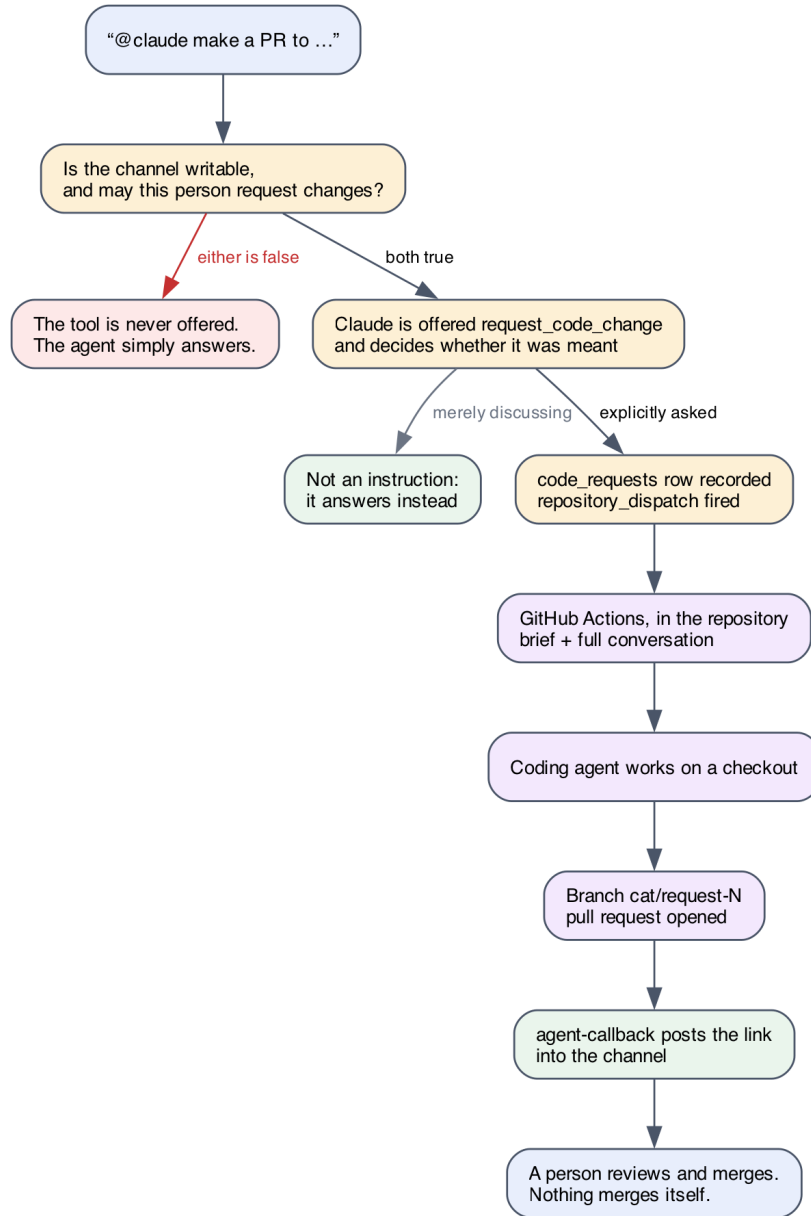


Figure 7: Asking for a change. The permission questions are settled before the model is given the opportunity, and the last step is always a person.